

CF-Verify: Counterfactual Evidence Deletion Pipeline

Inputs

(i) Self-rationale

(ii) Targeted deletion

(iii) Random deletion

(iv) Routing

Question Q

Answer call: $a(0) \sim p(.|Q, E)$

$E_- = E \setminus \hat{G}$

$R \sim \text{Unif}(E \setminus \hat{G})$

$\text{CFScore} = \hat{F}_T - \hat{F}_R$

Evidence E

Self-rationale: predict $\hat{G} \subseteq E$

$a(1) \sim p(.|Q, E_-)$

$a(2) \sim p(.|Q, E \setminus R)$

Route on τ

Merged variant: $a(0) + \hat{G}$ in 1 call

$\hat{F}_T = 1[a(1) \neq a(0)]$

$\hat{F}_R = 1[a(2) \neq a(0)]$

Accept / Revise / Re-Retrieve

Calls: 3+K (separate) or 2+K (merged). Cost: \$0.016/Q HotpotQA at K=1, \$0.030/Q at K=5.